

Linear algebra for ML: eight modules

1. Scalars

plain numbers and arithmetic

2. Vectors

components, dimension, types
magnitude, Euclidean distance
scalar and vector operations
dot product, angle, similarity
equation of a hyperplane
norms, vector spaces

3. Matrices

school level: types,
operations, inverse, determinant
intuition: rank, column space,
change of basis, linear
systems, linear transformations

4. Tensors

data of any shape as tensors

5. Eigenvalues, eigenvectors

the directions a matrix only
stretches (PCA)

6. Matrix factorisation

LU, QR, eigen-decomposition,
SVD

7. Advanced topics

quadratic forms (SVM)
Moore-Penrose pseudo-inverse

8. Tools

NumPy, SciPy